

Hadwiger's Conjecture

$\forall t \geq 0$, every graph is either t -colourable or it has a subgraph that can be contracted to K_{t+1}

$$\chi(G) \leq t+1$$

$\Rightarrow G$ contains K_{t+1} as a minor.

-1 direction $(\Rightarrow)$ not $(\Leftarrow)$.

$t \geq 4: \Rightarrow K_{t+1} = K_5$ (non planar)

$\Rightarrow$ if no K_5 minor, $(4=t)$ -colourable

< no loops, parallel edge graphs

$\Rightarrow$ simple

• contract edges of a graph G to form a subgraph is a minor

planar $\Rightarrow$ no $K_{3,3} / K_5$.

- every bipartite graph is 2-colourable
- complete (t) -partite graph has K_t
- an (t) -partite graph is ~~not necessarily~~ $\leq t$ -colourable

Let Hadwiger's Conjecture be denoted by $HC(t)$ be the statement that every graph w/ no K_{t+1} minor is t -colourable

- $HC(t)$ for $t \leq 3$ [1948]
- Hadwiger
 - $HC(6)$ is equivalent to $\chi \leq 5$
- Wagner 1987
-

~~the above~~

$\chi(\Gamma) \geq t+1 \Rightarrow \Gamma$ contain K_{t+1}

$\chi(\Gamma) \geq 2 \Rightarrow \Gamma$ contain K_2

$t=2$

$\chi(\Gamma) \geq 3 \Rightarrow \Gamma$ contain K_3

χ is the least colouring number

~~magner~~ pro Appel, Haken

$t=0$:

any graph w/ ≥ 1 node (k_1) has $\chi(G) \geq 1$

$t=1$:

$\chi(G) \geq t+1 = 2$,

graph has an edge, since no edges $\Rightarrow 1$ -colorable

$t=2$:

graph $\chi(G) \geq 3$ has K_3 minor.

graph

~~graph~~ G contains K_3 , cycle at least

length 3. $\Rightarrow$ not bipartite

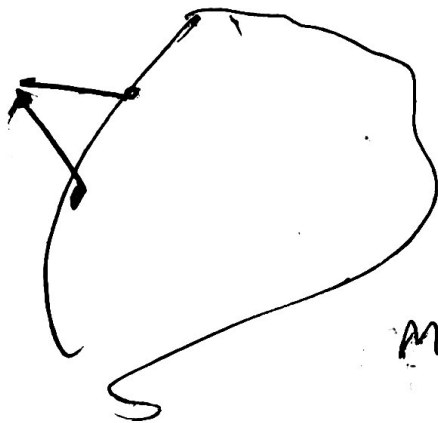


$t=3$:

$\chi(G) \geq 4 \Rightarrow$ has K_4 minor.

a graph w/ no K_4 minor is

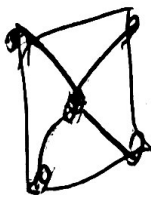
Induction



assume graph has no
K₄ minor, it is

3-colourable, if only
max degree is 3.

- use induction on graph to prove
3-colourability w/o K₄ minor



2 graphs G_1, G_2 and for $r=1, 2$

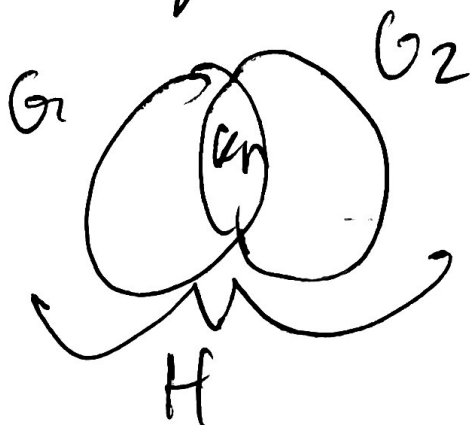
let C_r be a clique in G_r

clique: a subset of vertices all
pairwise adjacent (complete).

$|C_1| \geq |C_2|$ assumed.

check bijection between the cliques
and identify each vertex of C_1 w/
corresponding vertex of C_2 .

→ obtain graph H w/ 2 subgraphs
isomorphic to G_1 & G_2 overlapping on
the cliques.



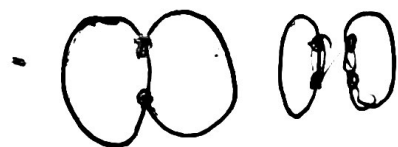
let G be a graph obtained from H ,
by deleting some edges, edges deleted
are of the cliques

G is a clique sum of G_1 & G_2
and if clique has size n , it is
called n -sum

* if G_1, G_2 t -colourable, H is t -
colourable

Theorems $\Rightarrow$ t -colourable proved for $H(G)$
for $a \leq t \leq B$, the graphs

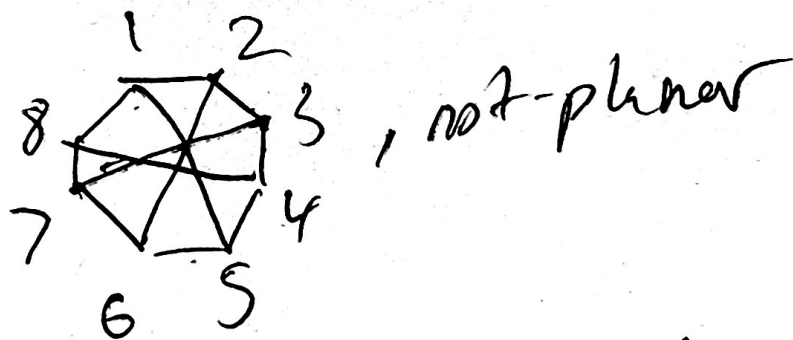
with no K_{t+1} minor are precisely the
graphs that can be built by repeated
clique-sums starting from graphs with at
most t vertices



Theorem (Wagner): 1937

The graphs with no K_5 minor are precisely the graphs that can be built by repeated 0, 1, 2, 3 ~~edge~~ sums, starting from planar graphs and copies of K_2 .

V_8 :



* an n -sum is a graph formed by cliques of size n .